

EmacEArt Cool Water — Quick Start

Stylized toon water for Godot 4: lakes, ponds, pools, roadside puddles. It goes on an ordinary plane placed over a dip in the terrain.

Everything you see on the surface — colour bands, shore glow, foam, light patterns on the bottom — comes from one number: the vertical distance between the surface and the ground below it. The water needs no textures or masks of its own.

What you need

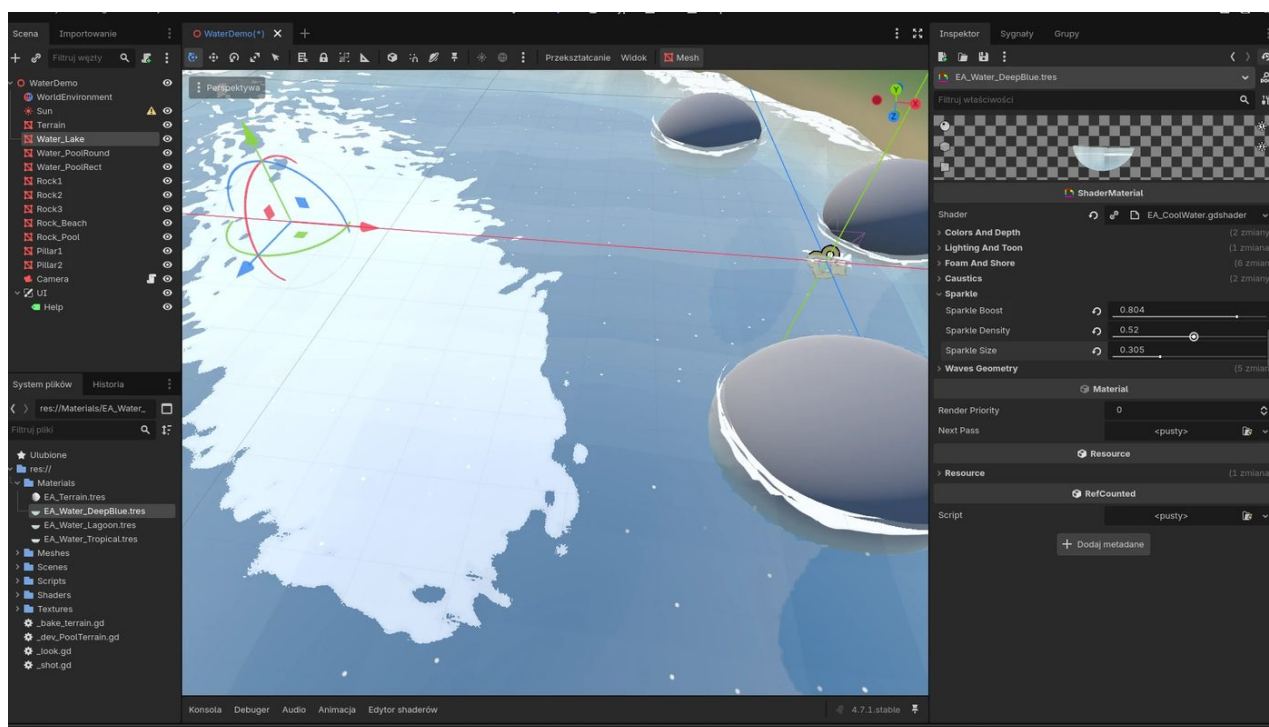
Godot **4.7 or newer**, renderer **Forward+ or Mobile**.

UWAGA. The Compatibility renderer will not work. Godot stores scene depth in a different format there, and this whole shader is built on reading the depth buffer.

Opening the demo scene

In the Godot project manager: **Import**, point at project.godot, **Import & Edit**. Then press F5, or open Scenes/WaterDemo.tscn and press F6.

The scene is finished the moment it opens. The terrain is a saved mesh, not something generated when the game starts — it looks the same in the editor as it does when running.



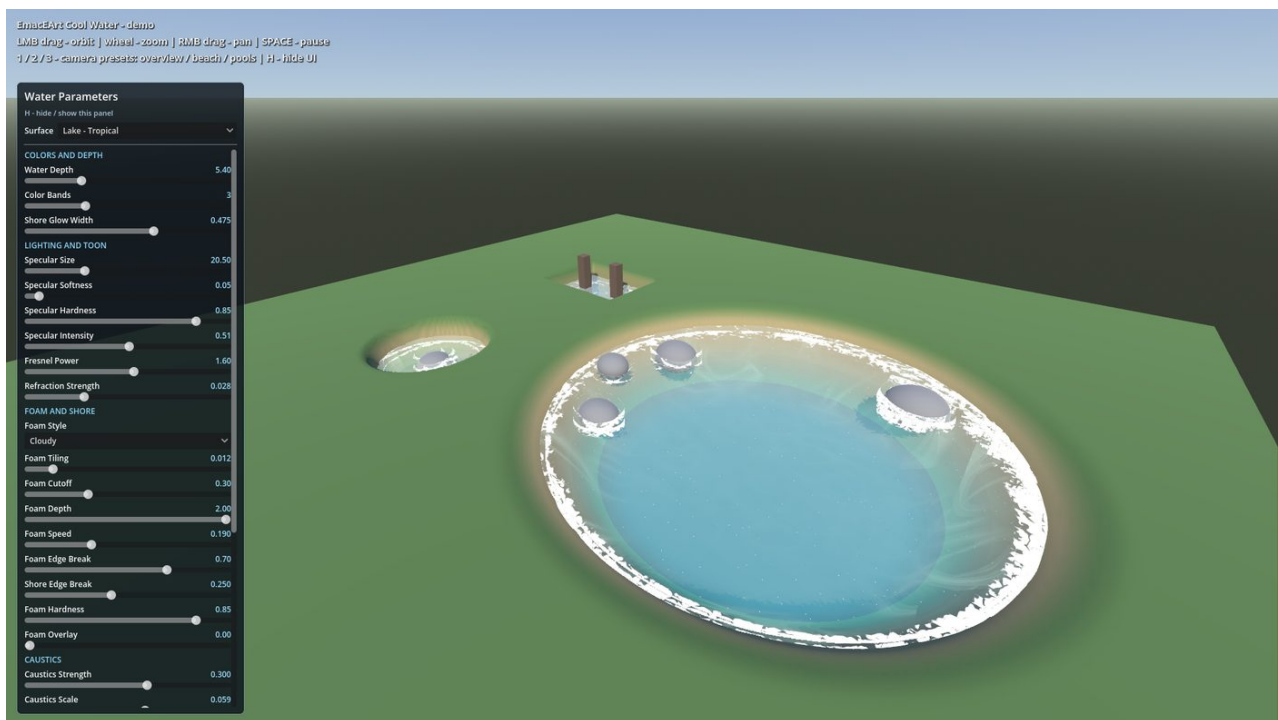
Demo scene open in the editor

Controls

Input	Action
LMB + drag	orbit the camera around the target
RMB + drag	pan the target
Mouse wheel	zoom
1 / 2 / 3	overview / beach level / both pools
Space	freeze time — waves, foam and caustics stop
H	hide and show the slider panel together with the help text

The slider panel

The panel appears on the left once you run the scene. Pick a surface at the top — lake, round pool or rectangular pool — and drag the sliders. The water reacts instantly.



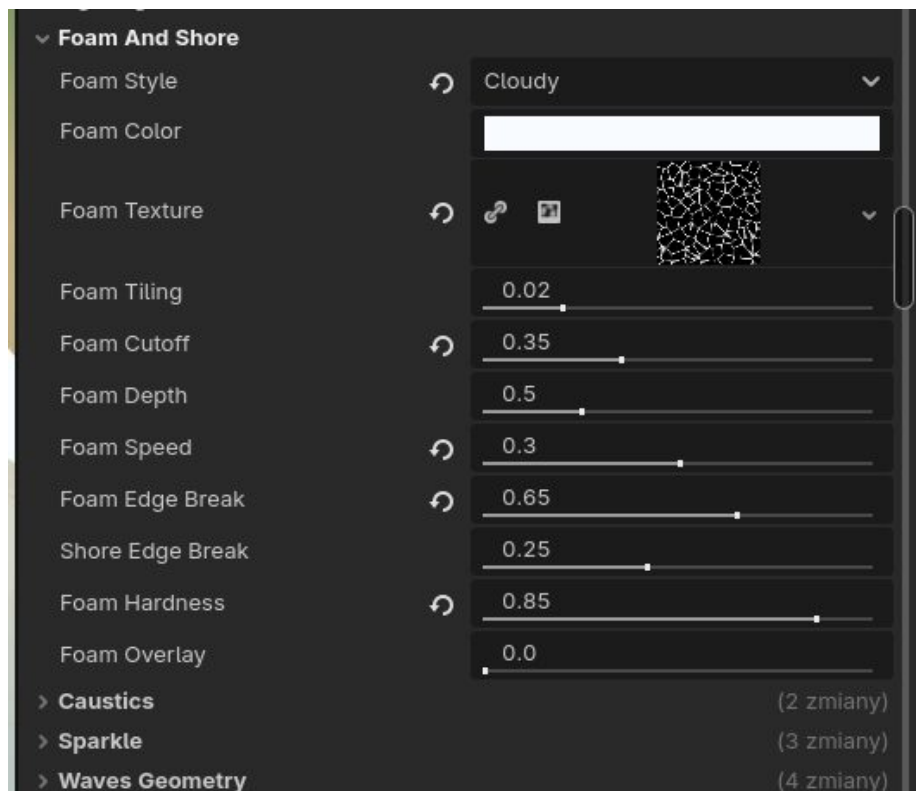
Slider panel in the running demo

Panel settings live only until you close the game. The .tres material files are never written to, so you can experiment without any risk to the project.

Press **H** before taking screenshots — it hides the panel and the help text.

Parameters in the editor

Select a water surface in the scene tree, then Inspector → **Material Override** → **Shader Parameters**. The parameters are split into six groups.



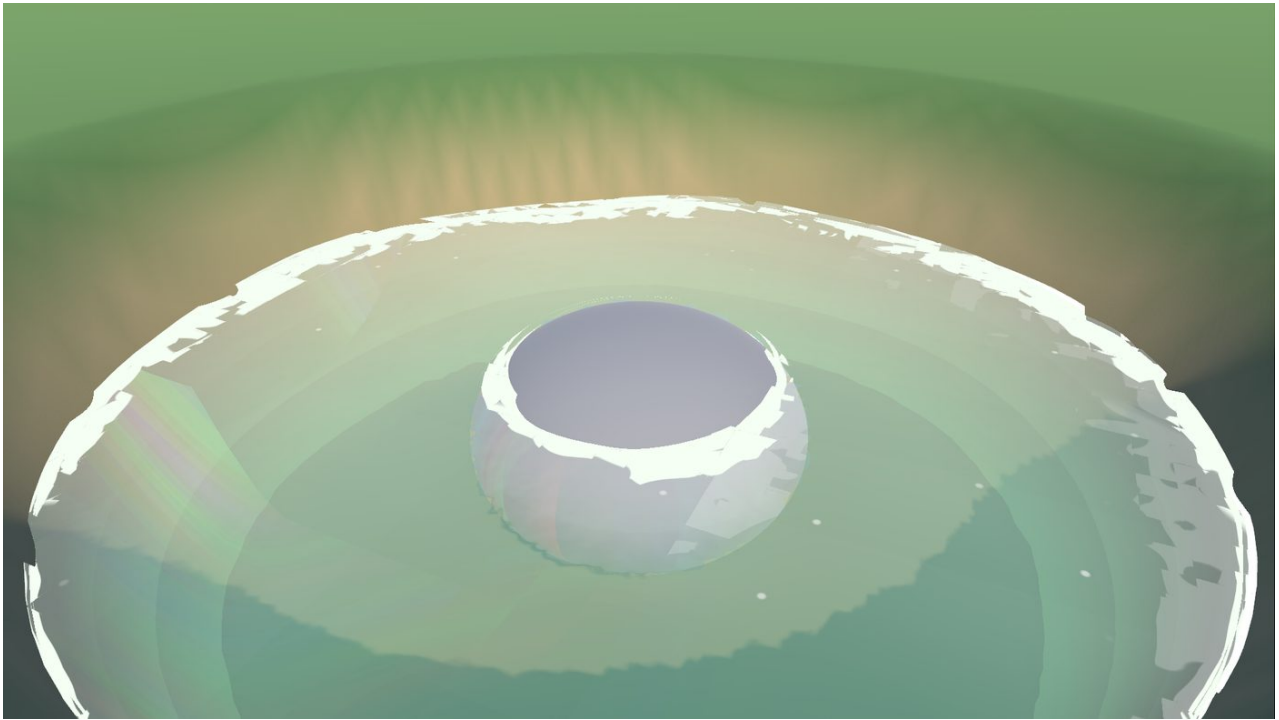
Foam and Shore group in the Inspector — the largest of the six, eleven foam parameters

Group	What it covers
Colors and Depth	shallow and deep colour, depth reference, band count, shore glow
Lighting and Toon	toon highlight and its flattening, fresnel, refraction strength
Foam and Shore	foam pattern, reach, edge break-up, hardness, blend mode
Caustics	tint, strength, scale and speed of the light veins on the bottom, chromatic aberration
Sparkle	flickering points on the surface — strength, density, size
Waves Geometry	normal map, its tiling and strength, animation speed, wave height

Water in your own scene

1. Add a MeshInstance3D with a **subdivided plane** — vertex waves need vertices to act on. 2. Assign one of the three materials from Materials/ as **Material Override**. 3. Place the plane over a dip in the terrain. There has to be geometry underneath — over empty space there is nothing to measure and the water goes flat.

Three ready materials ship with the pack: EA_Water_Tropical (turquoise shore water), EA_Water_DeepBlue (cool pool blue), EA_Water_Lagoon (green pond).



The round pool with the Lagoon material — the only one in the demo with chromatic aberration enabled

Further reading

The full documentation (Documentation_EN.md) covers all 27 parameters with their range and default, the three vertex colour channels (cutting the surface away, freezing the animation, painted foam), the three materials side by side, and a list of things better left alone in the project.

Join us on Discord

This pack keeps growing, and the next versions come out of what the people using it report back. On the server you can show how the water looks in your own scene, ask about settings, report a fault or request a parameter that is missing.

<https://discord.com/invite/ctXaf5Ftqw>

— EmacEArt